Algebra II with Trigonometry Chapter 5.2

Olympiad Solutions (Gemini)

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Solutions

1. An observer looking at the top of a radio tower measures an angle of elevation of 15°. After walking exactly 100 meters directly toward the tower on level ground, the angle of elevation increases to 30°. Mentally compute the exact height of the tower in meters.

Let A be the initial position, B be the second position, T be the top of the tower, and F be the base of the tower.

In $\triangle ABT$, the exterior angle at B is the angle of elevation $\angle TBF = 30^\circ$. By the Exterior Angle Theorem, this exterior angle is equal to the sum of the two opposite interior angles, so $\angle TBF = \angle BAT + \angle BTA$. Substituting the known angles gives $30^\circ = 15^\circ + \angle BTA$, which means $\angle BTA = 15^\circ$.

Because $\angle BAT = \angle BTA = 15^\circ$, $\triangle ABT$ is an isosceles triangle with sides $BT = AB = 100$ meters.

Now look at the right triangle $\triangle TBF$ formed by the tower. The height of the tower TF is the side opposite the 30° angle. Using the sine ratio: $TF = BT \sin 30^\circ = 100 \left(\frac{1}{2}\right) = 50$.

The exact height of the tower is 50 **meters**.

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2. Without using a calculator, determine the exact numerical value of the expression: $(\sin 15^\circ + \cos 15^\circ)^2$

Expand the binomial square to get: $\sin^2 15^\circ + \cos^2 15^\circ + 2\sin 15^\circ \cos 15^\circ$

Apply two fundamental trigonometric identities: the Pythagorean identity ($\sin^2\theta + \cos^2\theta = 1$) and the double-angle identity for sine ($2\sin\theta\cos\theta = \sin 2\theta$). This elegantly simplifies the expression to: $1 + \sin(30^\circ)$

Since $\sin 30^\circ = \frac{1}{2}$, the exact value is: $1 + \frac{1}{2} = \mathbf{\frac{3}{2}}$

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3. Given that $\sec\theta + \tan\theta = 5$ for an acute angle θ, mentally evaluate the exact value of $\sec\theta - \tan\theta$.

Recall the Pythagorean identity relating secant and tangent: $\sec^2\theta - \tan^2\theta = 1$

By factoring this as a difference of squares, we establish a direct relationship between the sum and difference of the two functions: $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$

Substitute the given value $\sec\theta + \tan\theta = 5$ into the equation: $(\sec\theta - \tan\theta) \cdot 5 = 1$

Dividing by 5, we immediately find: $\sec\theta - \tan\theta = \mathbf{\frac{1}{5}}$

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4. Exploit the symmetries of the right triangle to evaluate the following sum entirely in your head:
$\cos^2 10^\circ + \cos^2 20^\circ + \cos^2 30^\circ + \cos^2 40^\circ + \cos^2 50^\circ + \cos^2 60^\circ + \cos^2 70^\circ + \cos$

By the complementary angle identity (which reflects the symmetry of the two acute angles in a right triangle), $\cos(90^\circ - \theta) = \sin\theta$. Therefore, $\cos^2(90^\circ - \theta) = \sin^2\theta$.

We can pair the terms from the outside inward, matching each angle with its complement:
$(\cos^2 10^\circ + \cos^2 80^\circ) + (\cos^2 20^\circ + \cos^2 70^\circ) + (\cos^2 30^\circ + \cos^2 60^\circ) + (\cos^2 40^\circ$

Applying the complementary identity transforms each pair into the Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$. For example, the first pair becomes $\cos^2 10^\circ + \sin^2 10^\circ = 1$.

Since there are exactly 4 pairs and each pair evaluates to 1, the sum is simply: $1 + 1 + 1 + 1 = \mathbf{4}$

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5. Using the definitions of the trigonometric ratios, find the exact value of the following 89-term product:
$\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \cdots \tan 88^\circ \cdot \tan 89^\circ$

Recall the complementary angle identity for tangent: $\tan(90^\circ - \theta) = \cot\theta$. Furthermore, by definition, tangent and cotangent are reciprocals, so $\tan\theta \cdot \cot\theta = 1$.

We exploit this by pairing terms equidistant from the ends of the sequence: $\tan 1^\circ \cdot \tan 89^\circ = \tan 1^\circ \cdot \cot 1^\circ = 1$ $\tan 2^\circ \cdot \tan 88^\circ = \tan 2^\circ \cdot \cot 2^\circ = 1$ ...and so on.

There are 89 terms in the product. Pairing them leaves exactly one unmatched term in the exact center: $\tan 45^\circ$.

The entire product simplifies to 44 pairs multiplying to 1, times the middle term. Since $\tan 45^\circ = 1$, the product is: $1^{44} \cdot 1 = \mathbf{1}$

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6. In a right triangle with acute angles α and β, it is known that $\log_2(\sin\alpha) + \log_2(\cos\alpha) = -2$. Mentally determine the degree measures of both α and β.

Using the sum-to-product property of logarithms, combine the left side: $\log_2(\sin\alpha\cos\alpha) = -2$

Converting this logarithmic equation to exponential form yields: $\sin\alpha\cos\alpha = 2^{-2} = \frac{1}{4}$

Multiply both sides by 2 to construct the double-angle identity for sine ($2\sin\alpha\cos\alpha = \sin 2\alpha$): $\sin(2\alpha) = \frac{1}{2}$

Because α is an acute angle in a right triangle, $0^\circ < 2\alpha < 180^\circ$. The angles in this domain with a sine of $\frac{1}{2}$ are 30° and 150°. Thus:
$2\alpha = 30^\circ \implies \alpha = 15^\circ$ $2\alpha = 150^\circ \implies \alpha = 75^\circ$

Because α and β are the two acute angles of a right triangle, they must be complementary ($\alpha + \beta = 90^\circ$). If $\alpha = 15^\circ$, then $\beta = 75^\circ$, and vice versa.

The degree measures of the two angles are exactly 15° **and** 75°.